Democratising AI Platform Engineering with Open Source

Own It, Run It, Trust It

AI Leaders in Tech Night · Lisbon

Agent

Introduce our speaker to the room



Meet **Alfonso “Poncho” Sandoval**.

Developer Advocate and open-source maintainer at **Cisco Systems**, based in **Lisbon, Portugal**.

Today he shows how to own it, run it and trust it.

GitHub: ponchotitlan LinkedIn: asandovalros

I want agents to be part of my Monday

Not a moonshot. Someone to triage the ticket, check the logs, draft the reply and find the runbook, while your team gets on with the real work.

Everyday means every team, every day. That is where it gets hard.

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Count the moving parts

01 🧠 Models

02 🤖 Agent framework

03 🎼 Orchestration

04 🎯 Skills

05 🔀 Workflows

06 🔧 Tools and MCP servers

07 🔌 Integrations

08 📚 Knowledge bases

09 💭 Agent memory

10 🪪 Identity and access

11 🔑 Secrets

12 ✅ Approvals

13 🔭 Observability

14 ⏰ Triggers and schedules

15 🚀 Deployment

16 🖥️ Web UI

16 parts. Each one is a decision someone has to own.

A working list, not exhaustive.

What is AI platform engineering?

The infrastructure behind running agents and their tools safely:
authenticated, **authorized** and **traceable**.

Without it, every team solves access, tools, knowledge and approvals on its own.

With it, those decisions are made once, in the open, and every agent inherits them.

The onboarding checklist for an agent

1 🪪 **Who can use it?** — An access matrix

2 🔧 **What can it touch?** — A short tool list

3 📚 **What does it know?** — Knowledge bases

4 ✅ **Who approved it?** — Human in the middle

5 🔭 **What happened?** — A trace

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Three options

Build it yourself

YOU GET total control.

YOU CARRY every part, and its upkeep.

Buy a proprietary platform

YOU GET a fast start.

YOU CARRY answers you cannot inspect.

Adopt open source

YOU GET a running start, and source code you can inspect and adapt.

YOU CARRY running it, and fixing it when it breaks.



I Inês & T Tiago

I

Inês

Tiago, your project is awesome! Would you consider open sourcing it?

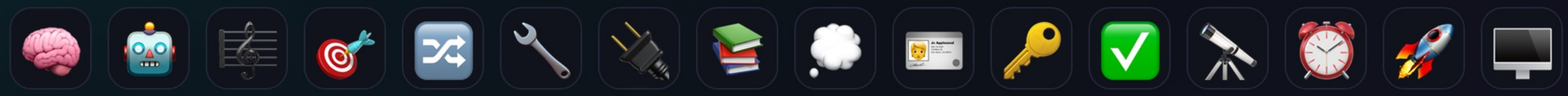
Tiago

Yes, sure! This is the project:

“Build me a popular open-source project called superflow. Make no mistakes.”

T

You don’t have to reinvent the 16 parts



Prompt it yourself

Generate all 16 parts, one by one. Then **own every one of them**: the bugs, the upgrades and the security review.

Adopt open source

They already exist, and the five questions are already answered. You get **source code you can inspect, tinker with and adapt**, plus a community that fixes it.

Spend your prompts on your problem, not on the platform.

Why open-source AI platforming, for a leader

The platform that runs your agents and their tools is the one you need to trust the most.

No lock-in

Your agents, skills and knowledge bases stay yours if a vendor changes terms or pricing.

Trust you can audit

Inspect how the platform handles access, approvals and traces for every agent, instead of taking it on faith.

You can tinker with it

Adapt the platform to your specific needs: your tools, your workflows, your rules. No waiting for a vendor roadmap.

You can give back

Contribute improvements and features born from your experience in the field, like issue #3090, and the whole community benefits.

The price: you run it. For an AI platform, that is worth knowing up front.
Mind your bandwidth: if your engineering team does not have the capacity to run it, a managed option may fit better.

An open-source alternative worth considering

Meet **CAIPE**, pronounced “cape”. Just as a cape empowers a superhero, it empowers teams with agentic AI automation.

CNCF Incubating project | Apache 2.0 | Self-hosted | Secure from day one

One answer, not the answer.



WHAT CAIPE.IO SAYS

Open Source AI Platform for All

Build, govern, and operate secure AI agents and workflows for **teams, enterprises, individuals, you**.

What is inside, and what you can swap

The open-source parts of CAIPE, grouped by purpose. The highlighted ones can be swapped if your stack differs.

🧠 Agents and state

Dynamic agents

Autonomous agents

Models ⇄
Claude, OpenAI, Gemini or any compatible endpoint

Document store ⇄
MongoDB or DocumentDB

🔧 Tools

MCP tool servers ⇄
GitHub, Jira, Confluence, ArgoCD, PagerDuty, Komodor, Backstage, Splunk, Slack, Webex, SharePoint, NetUtils. Bring your own.

📚 Knowledge

RAG server

Vector store
Milvus

Graph store
Neo4j, optional

Ingestors ⇄
Web, Jira, Confluence, Slack, Webex

🔐 Identity and access

Keycloak ⇄
OIDC single sign-on, Okta supported

OpenFGA
Who may do what

agentgateway
Gate for tool calls

🔭 Observability

Langfuse
Traces, tokens and cost

Metrics ⇄
Prometheus and OpenTelemetry, shipped anywhere. We used Splunk.

💬 Front doors

Web UI

Chat CLI

Slack bot ⇄

Webex bot ⇄

Swappable ⇄ **Default stack** From CAIPE's docker-compose and caipe.io. CAIPE is Apache 2.0, and each component keeps its own licence.

The scenario: LedgerLite

LedgerLite
A fictional invoicing and card-payments app, running on Kubernetes

25 minutes ago
A new release went out

Now
Paying an invoice fails

WHAT HAPPENS NEXT

1 **Indira**
Payment fails, she reports it

2 **Jira ticket**
Created in the shared project

3 **Beatriz**
Runs the Incident Copilot

4 **Agents**
Cluster Agent and Knowledge Agent investigate

5 **Diagnosis**
Posted on the ticket

6 **Beatriz**
Approves or rejects

PEOPLE

Indira
Customer

João
Support

Beatriz
On-call engineer

Miguel
Owner

AGENTS

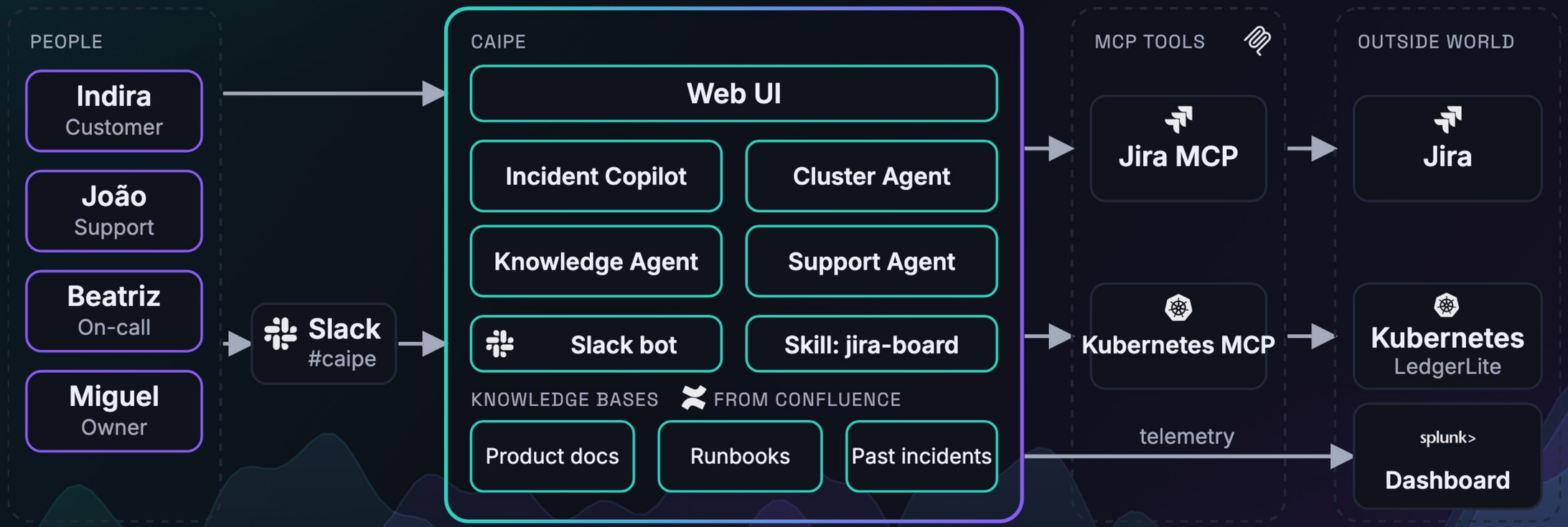
Incident Copilot
Resolves a ticket end to end

Cluster Agent
Investigates the cluster

Knowledge Agent
Answers from docs, with citations

Support Agent
Writes customer-safe replies

What is in the picture



The agents do not change, only the front door: Slack reaches the same agents, teams and approvals as the web UI.

Lessons from the field

A fresh install that failed

An image that would not pull, a service that exited at once. 7 issues filed on the CAIPE repository so far.

LESSON Budget time to run it, and file issues as you go.

A Slack user with zero tools

Created from Slack alone, they got a 403 from every tool server until a first web UI sign-in.

LESSON Identity is part of the platform. Test as a real person.

FROM LESSON TO CONTRIBUTION

Six traps, one proposal

Putting CAIPE behind my own domains and a tunnel cost me six traps. One service restarted about **5,900 times** over a single setting.

So I proposed **#3090**, an official guide to running CAIPE on custom domains:

- every URL and issuer setting, public or internal
- one place to set the domain, not eight
- a checklist to verify it works

LESSON Turn your scars into a proposal.

Forge at Splunk

One Splunk team: the Splunk Cloud Platform organization.

- 👨‍💻 **300+** engineers within a month of launch
- 💬 **90+** Slack channels
- 🤖 **About 32%** resolved with no human engagement
- ⏱️ **3.7 hours** baseline to “mere minutes”

Source: Outshift by Cisco blog, published 2026-04-24. Figures self-reported by the team.

Your next Monday

Start small. Take the five questions to your next review.

STEP 1

Own it

Pick one painful workflow and an open-source platform you can **inspect, tinker with and adapt**.

STEP 2

Run it

Start **read-only**: agents look, never touch. Watch adoption, cost and latency.

STEP 3

Trust it

Add **approvals**, so a human says yes first. Then let agents act, with every action on record.

1 **Who can use it?**

2 **What can it touch?**

3 **What does it know?**

4 **Who approved it?**

5 **What happened?**

No answers to the five questions, no next step.

Questions

Scan a code, or click a title, to keep going.

START HERE

🌐 The official CAIPE website

caipe.io

CONNECT WITH THE SPEAKER

💼 LinkedIn

linkedin.com/in/asandovalros

RUN IT YOURSELF

🧪 The session demo on GitHub

github.com/ponchotitlan/ai-platform-for-the-many-caipe

LEARN IT STEP BY STEP

🎓 CAIPE Learning Lab on DevNet

developer.cisco.com/learning